

AN INTERNSHIP REPORT ON

Industrial Automation Training at B&R Industrial Automation

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
IN THE FULFILLMENT OF THE REQUIREMENTS
FOR THIRD YEAR INTERSNHIP EVALUATION

**THIRD YEAR OF ENGINEERING (ELECTRONICS AND TELECOMMUNICATION
ENGINEERING)**

SUBMITTED BY

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SAVITRIBAI PHULE PUNE UNIVERSITY

Internship Report
On
Industrial Automation Training
At
B&R Industrial Automation

Indira College of Engineering and Management, Pune
Department of Electronics and Telecommunication Engineering



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ACKNOWLEDGEMENT

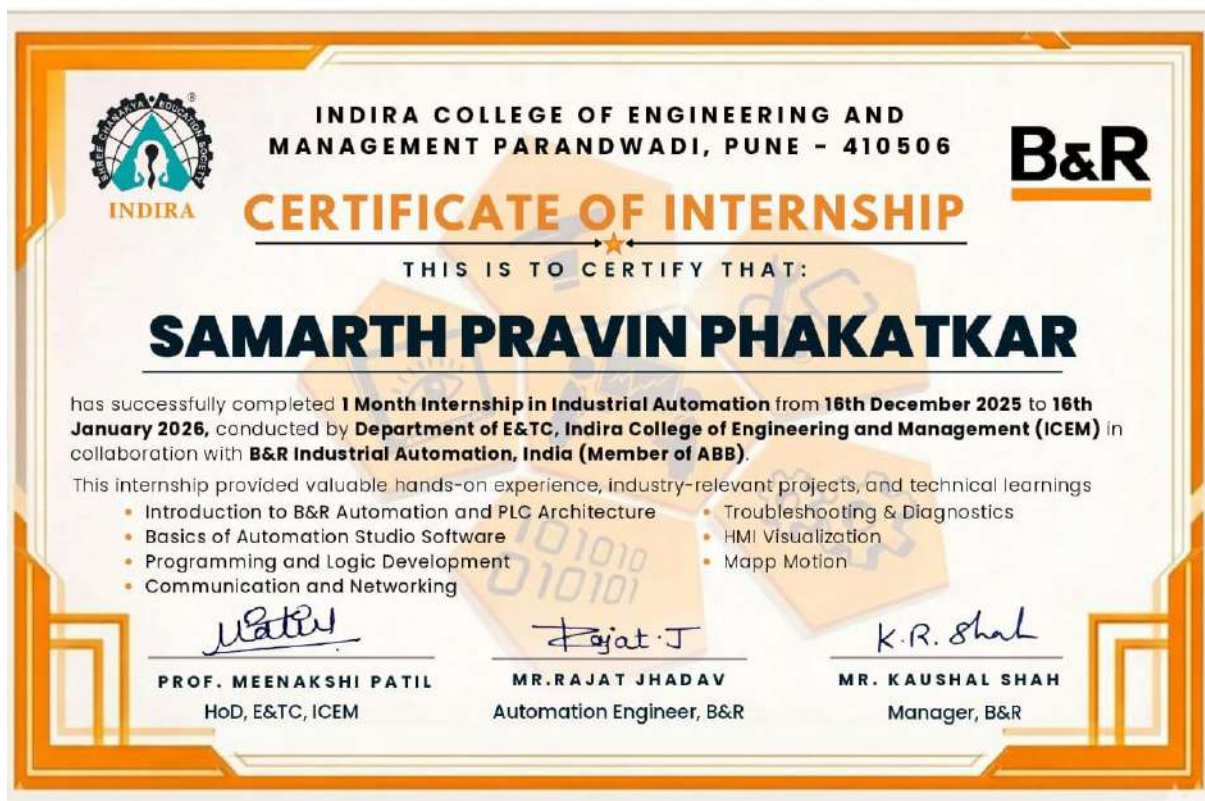
This report has been prepared based on the internship completed at **B&R Industrial Automation**, a remote-based organization that provides online programs for students to gain practical exposure and understand the real-world implementation of theoretical concepts. The primary objective of this internship was to fulfill the academic requirements of the course (310255: Internship) while also developing industry-relevant skills. Through this program, I was able to bridge the gap between classroom learning and practical application in the field of web development.

The internship offered a structured learning environment where I could explore various aspects of frontend development, work on assigned tasks, and enhance my technical knowledge. Being a remote internship, it also helped me adapt to a professional work culture in a virtual setting, improving my communication skills, self-discipline, and ability to work independently.

I would like to express my sincere gratitude to HOD Prof. Meenakshi Patil, Prof. Sudhir Sawarkar Internal Mentor, Internship Coordinator Prof. Priyanka Patil and ENTC department for providing me with the opportunity to undertake this internship and for his constant guidance and support throughout the duration of the program. I am also thankful to the team at **B&R Industrial Automation**, including mentors, coordinators, and staff members, for their encouragement, valuable feedback, and assistance whenever required. Their support played a crucial role in helping me successfully complete the internship and gain meaningful learning experiences.

I am deeply grateful to everyone who contributed directly or indirectly to the successful completion of this internship. Their guidance and motivation have helped me enhance both my technical and professional skills, which will be beneficial for my future career.

Internship Certificate



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1. INTRODUCTION

The internship at **B&R Industrial Automation** was undertaken to gain practical exposure to the field of **industrial automation**, which is one of the most important domains in modern engineering industries. Automation has become a crucial part of industrial processes because it helps improve efficiency, accuracy, safety, and productivity.

The main purpose of this internship was to understand the working principles of industrial automation systems and to study how automation technologies are applied in real-world industrial environments. During this internship, I got an opportunity to learn about various automation components and concepts such as **Programmable Logic Controllers (PLC)**, **Human Machine Interface (HMI)**, **industrial sensors**, **control systems**, and **automation software tools**.

Industrial automation is widely used in manufacturing, packaging, process industries, robotics, and smart production systems. As an Electronics and Telecommunication Engineering student, this internship helped me connect theoretical knowledge with practical industrial applications. It also gave me an understanding of how automation systems are designed, programmed, and controlled for different industrial tasks.

The training conducted during the internship provided both conceptual and application-based learning. It helped me understand how automation improves industrial performance and how modern industries rely on intelligent control systems for smooth operation.

Overall, this internship was a valuable learning experience that helped me strengthen my technical understanding, improve my industry awareness, and gain confidence in automation-related technologies.

2. PROBLEM STATEMENT

In many residential, commercial, and industrial buildings, lift/elevator systems are an essential part of vertical transportation. Traditional lift systems require efficient control, proper floor selection, smooth movement, door operation, and safety management. Manual or poorly designed systems may result in operational delays, incorrect floor handling, reduced efficiency, and safety concerns.

The Lift Project carried out during the internship aimed to understand and implement the automation logic behind elevator operation using industrial automation concepts. The project focused on how a lift can automatically respond to floor requests, move in the correct direction, stop at the selected floor, and operate in a controlled and safe manner.

The main problem addressed in this project was to design and understand a system that can automate lift movement and floor control using industrial control methods. This required understanding of control logic, input-output operations, sensors/switches, automation software, and sequential operation.

The project helped in studying how industrial automation can be used to create a smart, reliable, and efficient lift control system. It also gave practical knowledge of how real-life automation projects are planned and implemented using PLC and automation tools.

Thus, the problem statement of this internship was to study and implement an automated lift control system using industrial automation concepts.

3. OBJECTIVES

The main objectives of this internship were as follows:

- To understand the basics of industrial automation systems.
- To study the working of an automated lift/elevator control system.
- To understand how PLC-based control logic is used in lift operation.
- To learn how floor selection and directional movement of the lift can be controlled.
- To understand the role of inputs, outputs, switches, sensors, and relays in automation projects.
- To study the use of HMI and automation software in monitoring and control.
- To gain practical exposure to industrial automation project implementation.
- To understand sequence-based control in real-time systems.
- To improve technical knowledge in the field of automation and control engineering.
- To bridge the gap between theoretical learning and industrial application.

4. ACTIVITIES DURING INTERNSHIP

During the internship at B&R Industrial Automation, I was involved in various technical learning activities related to industrial automation and the implementation of a Lift Automation Project.

At the beginning of the internship, I was introduced to the fundamentals of industrial automation and its applications in modern industries. I learned how automation is used to control machines and processes efficiently with minimum human intervention.

A major part of the internship was focused on the Lift Project, where I studied the automation logic required for the operation of a lift system. I understood how a lift responds to user inputs such as floor selection and how it moves accordingly between different floors.

During the project, I learned how input and output devices play an important role in lift operation. Inputs such as floor request buttons, call switches, and control signals are used to instruct the system, while outputs such as motor control, lift movement indication, and floor display are controlled based on the logic.

I also studied how PLC logic can be used to control the movement of the lift in upward and downward directions. I understood how the system decides the lift position, floor stopping condition, and sequencing of operations. This helped me gain practical understanding of sequential control systems.

Another important part of the internship was learning about Human Machine Interface (HMI) and how it can be used to display lift status, floor information, and operational controls. This improved my understanding of how user interaction and monitoring are handled in automation systems.

I was also introduced to Automation Studio and related industrial software tools used in programming and system design. Through this, I got exposure to how industrial automation projects are designed, configured, and tested.

Overall, the internship activities were very informative and helped me understand how industrial automation concepts can be applied in a practical project such as an automated lift system.

5. TECHNOLOGY DETAILS, TOOLS AND SOFTWARE'S USED

During the internship and Lift Project, several industrial automation technologies, tools, and components were studied. These were used to understand and simulate the working of an automated lift control system.

Technologies and Concepts Learned

- Industrial Automation
- Lift / Elevator Control Logic
- PLC Programming Concepts
- HMI Basics
- Sequential Control Systems
- Input / Output Interfacing
- Industrial Sensors and Switches
- Automation-based System Monitoring

Tools / Software Used

- Automation Studio – Used for automation logic understanding, programming, and system design.
- PLC (Programmable Logic Controller) – Used for controlling lift logic and sequential operations.
- HMI (Human Machine Interface) – Used for displaying system status and floor control interface.
- Switches / Push Buttons – Used as floor request and control inputs.
- Relays / Control Devices – Used for switching and control actions.
- Training Setup / Simulation Setup – Used for understanding practical implementation of automation systems.

Technical Explanation of Lift Project

1. PLC in Lift Control

- The **PLC** acts as the main controller of the lift system. It receives input signals such as floor selection requests and then processes the logic to move the lift in the required direction. It also controls stopping conditions and floor response.

2. Input Signals

- The input section includes:
 - Floor request buttons
 - Call buttons
 - Position signals

3. Output Signals

The output section includes:

- Lift motor movement
- Up/Down indication
- Floor stopping control
- Lift status display

These outputs are controlled based on the PLC logic.

4. Sequential Logic

Lift operation follows a sequence. The system must:

1. Receive floor request
2. Compare current floor with selected floor
3. Move lift in correct direction
4. Stop at required floor
5. Wait for next command

This sequence helped in understanding **real-time control logic**.

5. HMI Role

The HMI helps in visualizing the lift operation, displaying floor number, and giving operator/user control over the system.

Thus, the Lift Project provided practical knowledge of how industrial automation is used in real-life control systems.

In addition, the HMI improves system efficiency by offering **real-time feedback**, reducing manual intervention, and minimizing operational errors. It enhances safety by clearly indicating system faults or abnormal conditions.

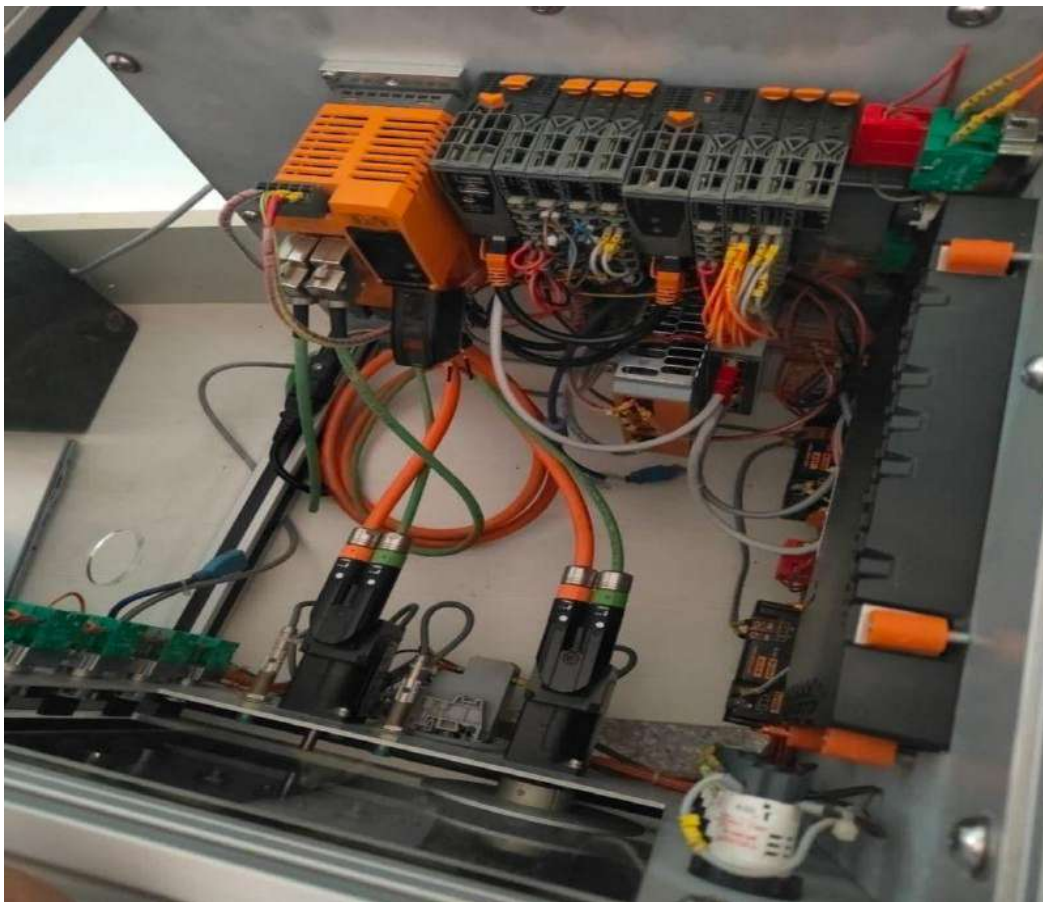
Thus, the Lift Project provides practical exposure to how **industrial automation systems integrate PLC and HMI** for efficient, user-friendly, and reliable control in real-life applications such as elevators and automated transport systems.

6. TRAINER KIT / EQUIPMENTS USED

- PL KIT AND DISPLAY HMI



- INSIDE PLC KIT CONNECTIONS



PLC TRAINER KIT EXPLAINED

◆ Overview

- A compact automation control panel used for training, testing, or small industrial processes
- Enclosed in a transparent protective cabinet for safety and visibility
- Integrated with Human Machine Interface (HMI) for monitoring and control

◆ Main Components

- PLC (Programmable Logic Controller) – Core controller for automation logic
- HMI Touch Panel – Used to interact, display data, and control operations
- I/O Modules – For connecting sensors and actuators
- Power Supply Unit (SMPS) – Provides regulated DC power
- Terminal Blocks & Wiring – Organized electrical connections
- Relays / Interface Modules – For switching and signal isolation

◆ Sensors & Actuators (Bottom Section)

- Photoelectric sensors / proximity sensors – Detect objects or position
- Pneumatic cylinders or linear actuators – For movement/operation
- Air tubes (orange pipes) – Indicate a pneumatic system
- Solenoid valves – Control air flow to actuators

◆ Features

- Real-time process monitoring via HMI
- Modular design – easy to modify and expand
- Safe operation due to enclosure and proper wiring
- Supports industrial communication protocols

◆ Applications

- Industrial automation training labs
- Conveyor or sorting system simulation
- Packaging / material handling demos
- PLC programming practice
- Used in colleges and labs to teach PLC programming and control systems.
- Simulates object detection, sorting, and movement.
- Demonstrates automation of production lines.

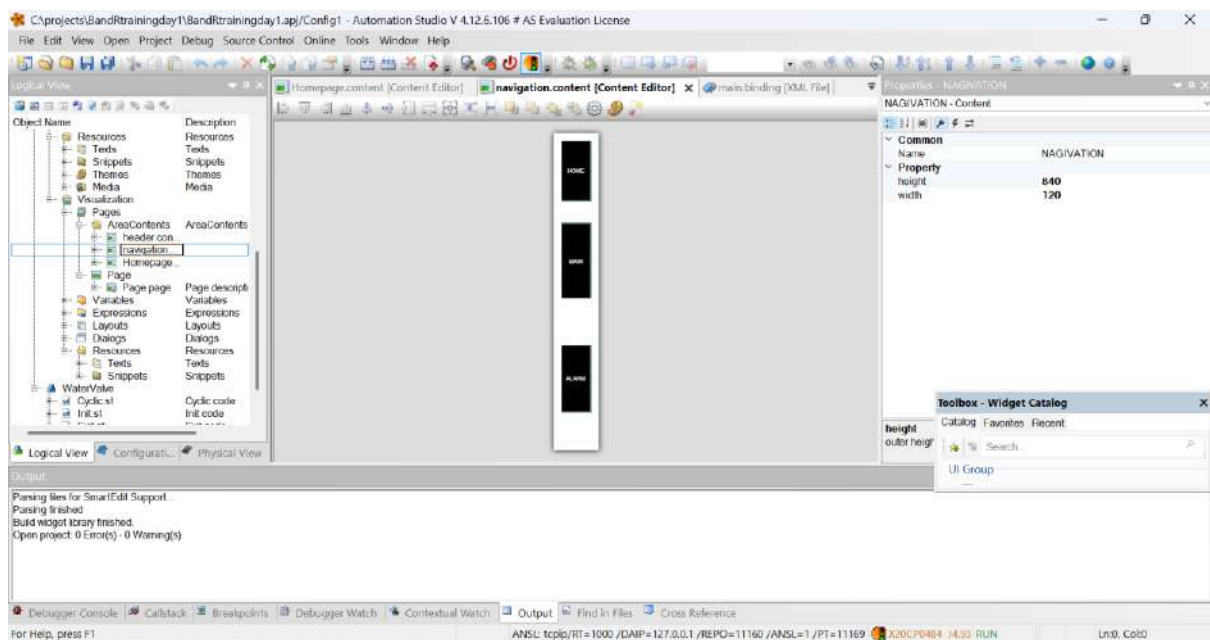
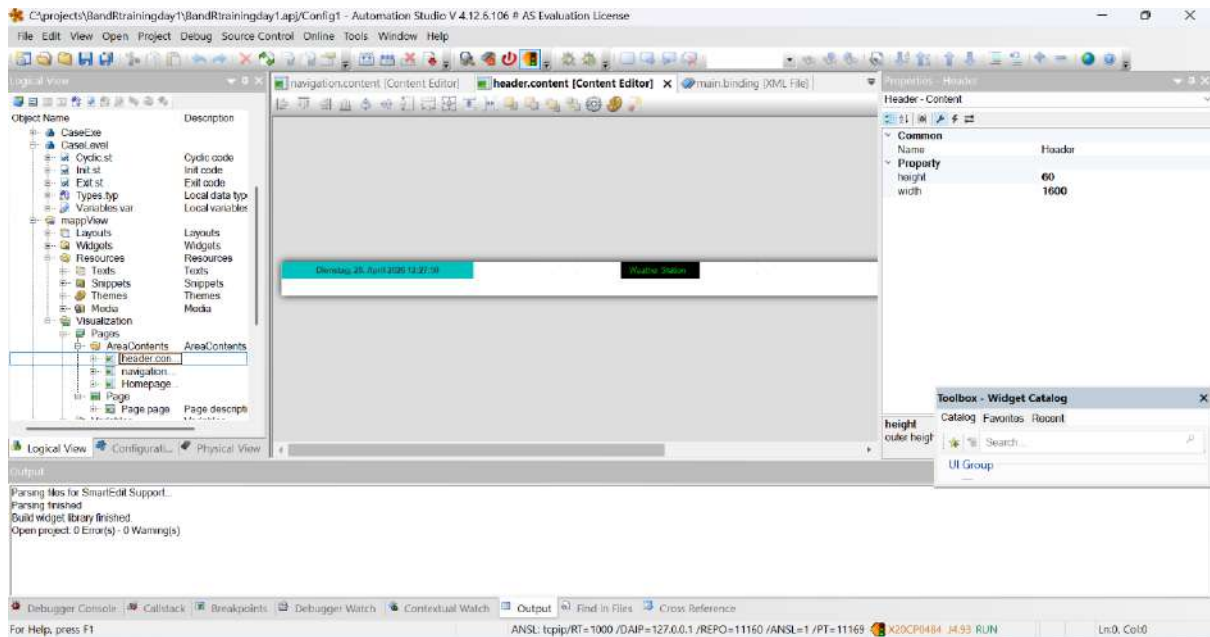
◆ Advantages

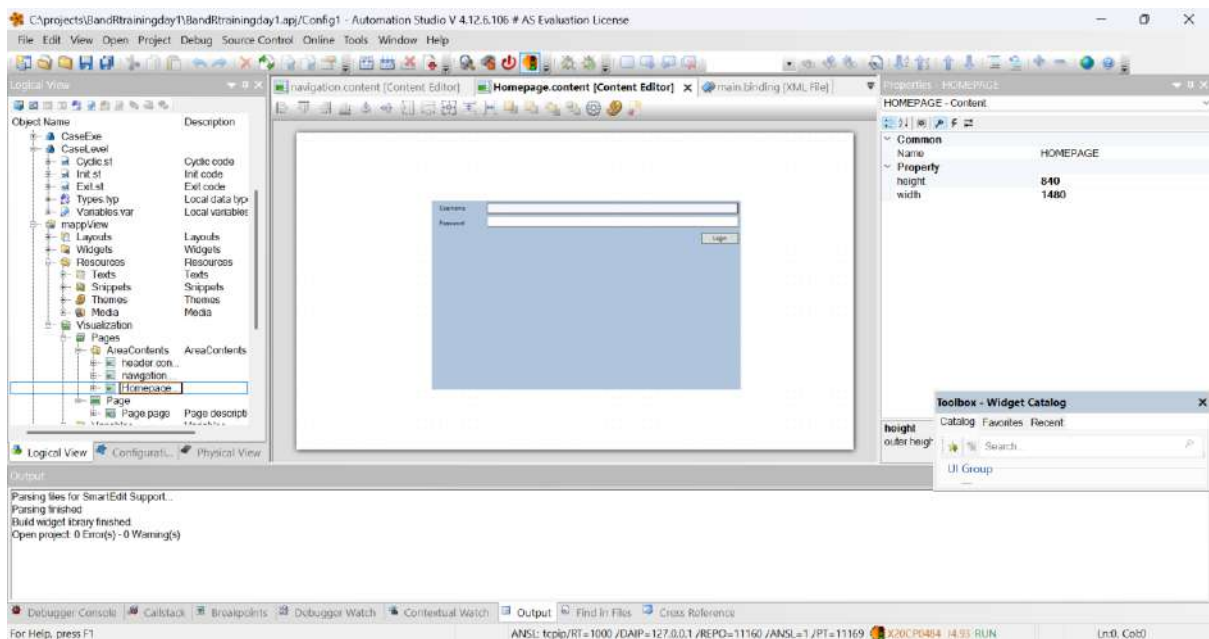
- Hands-on learning for automation
- Compact and portable setup.
- Clearly organized wiring and modular components make fault finding simple.
- Allows real-time monitoring and control, similar to real industry systems.
- Can be used for conveyor control, sorting, counting, automation logic, etc.
- Designed for repeated use with durable industrial-grade components.
- Demonstrates real industrial components.
- Provides hands-on exposure to actual industrial components like PLC, sensors, and actuators.
- Enclosed transparent box protects users while allowing visibility of internal operations.

✗ Disadvantages

- High Cost: PLCs, HMIs, and modules are expensive.
- Complex Setup: Requires technical knowledge (programming + wiring).
- Maintenance Needed: Wiring faults or module failures can occur.
- Limited Scalability: Training kits may not handle large industrial loads.
- Power Dependency: Needs stable power supply.
- Troubleshooting Difficulty: Fault detection can be time-consuming.
- Bulky Design: Not very portable compared to small embedded systems.

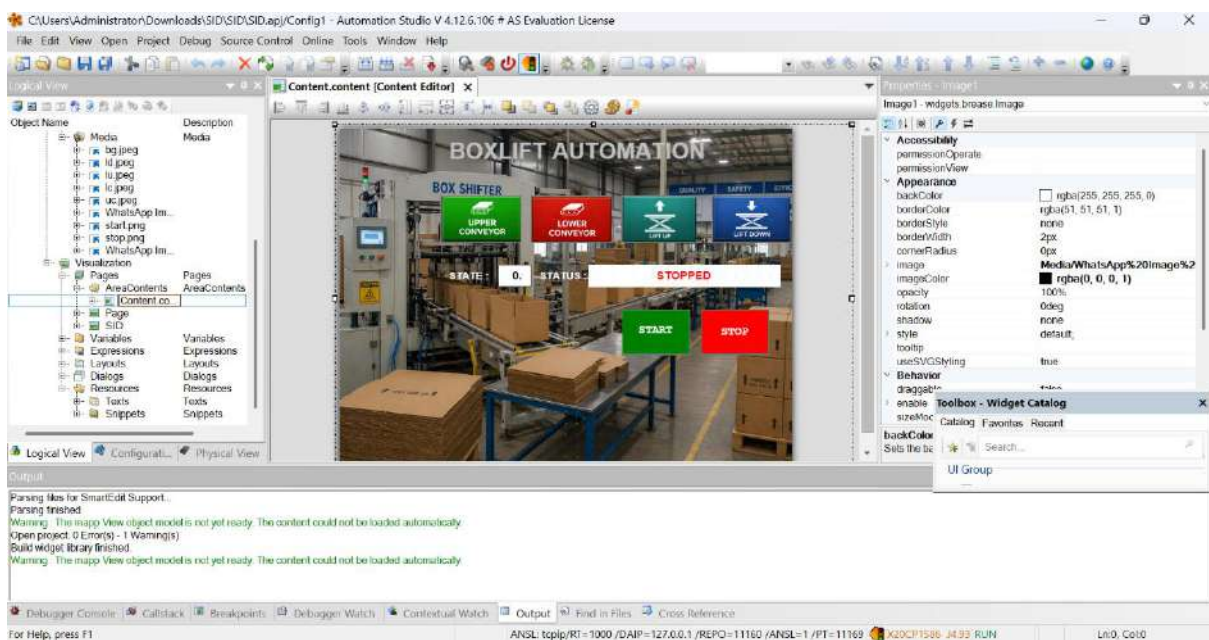
LEARNED IN INTERSHIP



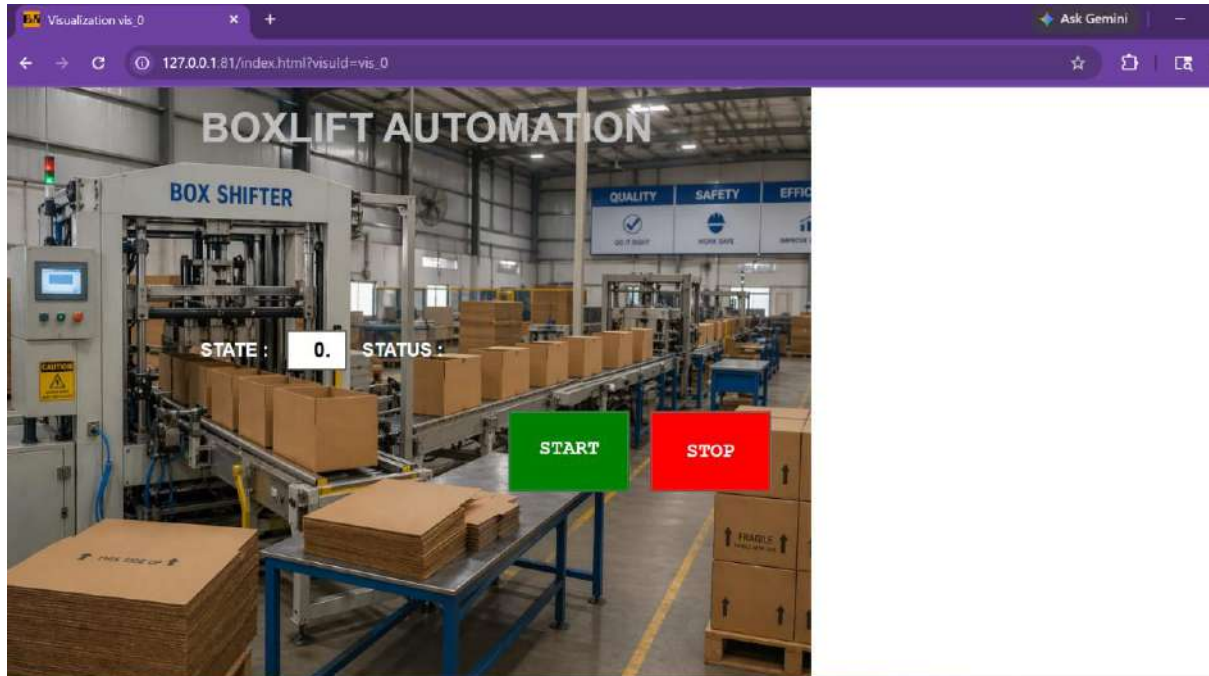


PROJECT (BOX LIFT)

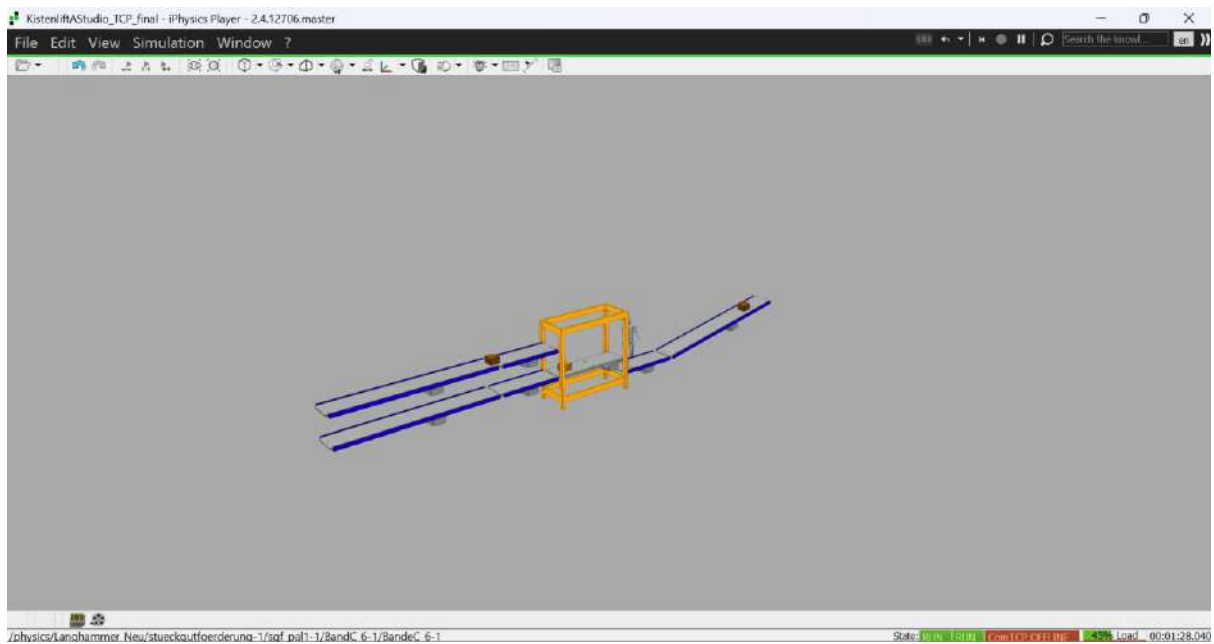
- USING AUTOMATION STUDIO



- USING THIS LINK OUTPUT OF PROJECT IS SHOW
- http://127.0.0.1:81/index.html?visuId=vis_0



- FINAL OUTPUT ON IPHYSICS



7. RESULTS AND CONCLUSION

The internship at B&R Industrial Automation was successfully completed and proved to be a valuable practical learning experience. Through this internship, I was able to understand the fundamentals of industrial automation and their implementation in a real project environment.

The main outcome of the internship was the understanding of the Lift Automation Project, in which I learned how automation logic can be used to control lift movement, floor selection, directional operation, and sequence handling.

Through this project, I gained practical knowledge of PLC-based control systems, input-output interfacing, automation software, and industrial project logic. I also understood how real-time systems are designed to operate efficiently, safely, and accurately.

The internship helped me improve my technical understanding of automation systems and gave me confidence in handling industrial concepts such as control logic, HMI, automation and control engineering.

In conclusion, this internship was very useful in bridging the gap between academic knowledge and practical implementation. The Lift Project helped me understand how automation can be used to solve real-world control problems efficiently and effectively.

8. REFERENCES

[1] Official Website

-  <https://www.br-automation.com/en/>
→ Overview of products, solutions, and industrial applications.

[2] PLC & Hardware Information

-  <https://www.br-automation.com/en/products/plc-systems/>
→ Details about PLC architecture, modular I/O, and control systems.

[3] Automation Studio (Programming Software)

-  <https://www.br-automation.com/en/products/hmi/>
→ Covers PLC programming, HMI design, communication, and diagnostics.

[4] Online Courses / Training

-  [B&R Online Courses \(Automation Academy\)](#)
→ Free & paid courses to learn PLC, HMI, motion control anytime.
-  [B&R Automation Academy Training Portal](#)
→ Classroom + virtual training programs for industrial automation.

[5] Tutorials & Learning Content

-  [B&R Tutorials & Instruction Videos](#)
→ Installation, maintenance, and system tutorials.

[6] Community & Support

-  [B&R Community Forum](#)
→ Ask questions, troubleshoot PLC/HMI issues, and learn from experts.

[7] Extra Info (Overview)

- B&R is part of **ABB Group** and provides:
 - PLC systems
 - HMI panels
 - Motion control (ACOPPOS drives)
 - Industrial PCs
- Known for integrated automation platform and high-speed control systems.